FLORIDA INTERNATIONAL UNIVERSITY

REGRESSION ANALYSIS ON WHAT VARIABLES AFFECT PENALTIES

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Abstract

Soccer is the most popular sport globally, and penalties are a game-changing role at most of the game. This paper chooses 75 independent professor soccer players and uses a linear regression model to see which variables are associated to penalties. I choose Jumping, Speed, Ball_Control, Balance, Heading, and Strength as independent variables. The final regression model will be trained from the data and used to predict penalties. Our final model uses only Ball_Control as explanatory variable to predict the Penalties and the model assumptions are satisfied.

Contents

1.Introduction

Using statistical models such as linear regression to analyze sports and improve athletes' ability in competition have been an interesting topic in recent years. Bangsbo, J., Mohr, M. and Krustrup, P. (2006) and Clemente, F. M., Couceiro, M. S., Martins, L., Manuel, F., Ivanova, M. O., & Mendes, R. (2013) are good examples of this area of research.

This paper will try to do a statistical investigation on whether a soccer player's penalty ability is associated with their other abilities such as jumping, speed, strength, and so on. As we know, the player's penalty ability is more comprehensive and is not easily predicted or explained compared to jumping, speed, heading, strength, and ball-control, etc. A multiple linear regression analysis is useful here to see how much jumping, speed, heading, balance, strength and so on can say about a player's penalty ability. Performance data from 75 independent soccer players was collected, comprising each player's average performance from January 2017 to January 2020. This dataset enables a clear overview of each player's soccer ability, as measured by their performance per season. This paper will build upon the chapter on Statistical Learning and Duke University's Data Analysis and Statistic Inference and the "R Language Battle" Return Model Chapter. In terms of data analysis, this paper will utilize a multiple linear regression model to analyze the dataset.

To implement the statistical procedure, a few common statistical software packages are used, including SPSS, R and EXCEL.

2.Data

2.1 Source of Data

This paper and its analytical method are based on publicly available soccer player online at https://www.kesci.com/home/dataset/5e79c46b98d4a8002d2cb73c/document. Descriptions of each variable used herein is shown in Table 1.

Variable	Explanation
Jumping	Athlete’s jumping ability, an integer of the value range 1 to 100, the greater the value, the stronger the ability.
Speed	Athlete's running speed, an integer of the value range 1 to 100, the greater the value, the stronger the ability.
Ball_Control	Athlete’s ability to control the ball, an integer ranging from 1 to 100. The greater the value, the stronger the ability.
Balance	Athlete’s balance ability, an integer of the value range 1 to 100, the greater the value, the stronger the ability.
Heading	Athlete’s heading ability, an integer of the value range 1 to 100, the greater the value, the stronger the ability.
Strength	Athlete’s strength ability, an integer of the value range 1 to 100, the greater the value, the stronger the ability.

Response Variable	Explanation
Penalty	Athlete's penalties, an integer of the value range 1 to 100, the greater the value, the stronger the ability.

2.2 Descriptive Statistics

We randomly select 75 individual players in the data set, which records the average performance of 75 soccer players from January 2017 to January 2020. The original data was provided by FIFA (Fédération Internationale de Football Association). The following is a simple descriptive analysis of the independent and dependent variables.

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Penalties	75	11	91	62.84	19.435
Jumping	75	34	95	73.43	11.798
Speed	75	33	96	74.27	12.933
Ball_Control	75	16	95	74.88	21.131
Balance	75	34	95	67.36	16.802
Heading	75	10	92	63.63	22.884
Strength	75	43	93	72.56	11.515
Valid N (listwise)	75				

Figure 1: Descriptive Statistics.

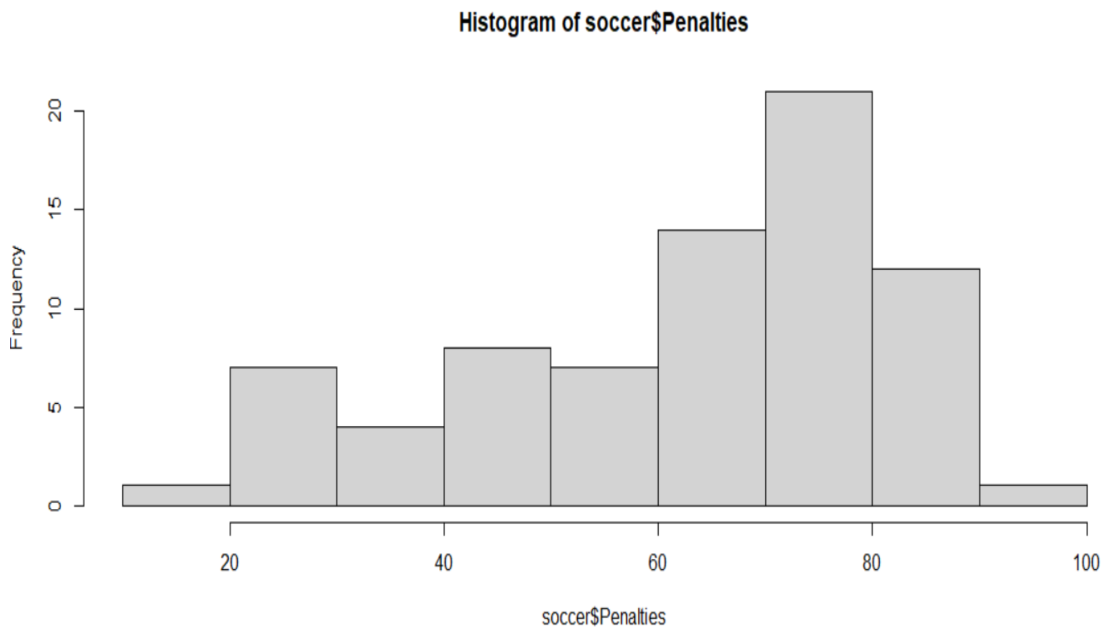


Figure 2. Histogram of the dependent variable.

2.3 Objectives of the Study

This research aims to investigate the linear association between the athlete's penalties and the other six abilities: jumping, speed, heading, balance, strength, and ball-control. A predictive model based on multiple linear regression will be built to predict penalties using these other abilities.

3.Fitting Linear Regression Model

3.1 Basic Theory of The Model

The basic form of the multiple linear regression model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p + \varepsilon$$

Where Y is the regression dependent variable, $X_i(i = 1,\ldots,p)$ is the regression independent variable,β_0 is the regression constant, $\beta_i(i = 1,\ldots,p)$ is the regression coefficient, ε is the random error, and there is $\varepsilon \sim N(0,\ \ \sigma^2 I_n)$.

Let $Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}, X = \begin{bmatrix} 1 & \cdots & x_{p1} \\ \vdots & \ddots & \vdots \\ 1 & \cdots & x_{pn} \end{bmatrix}, \beta = \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_p \end{bmatrix}$, where $y_i(i = 1,\ldots,n)$,

$$(x_{1i}, \ldots, x_{pi})(i = 1,\ldots n),$$

respectively represent the observed value of $Y, (X_1, X_2, \ldots, X_p)$. Accordingly, the form of the multivariate linear model becomes:

$$Y = X\beta + \varepsilon$$

The next step is to estimate the coefficient β. This is achieved by utilizing the least squares method. The principle of least squares even minimizes the residual sum of squares. Generally, random errors are used to estimate residuals. The function obtained is thus:

$$Q(\beta) = (Y - X\beta)^T (Y - X\beta) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{1i} - \ldots - \beta_p x_{pi})^2$$

Further, the β value that makes the function reach the minimum $Q(\beta)$ is the $\hat{\beta}$ we want to achieve, namely:

$$\hat{\beta} = \underset{\beta}{argmin} Q(\beta)$$

Resultantly, the final value of $\hat{\beta}$ is:

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

In this way we get the final regression equation:

$$\hat{Y} = X\hat{\beta} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \ldots + \hat{\beta}_p X_p$$

Let $\hat{\varepsilon} = Y - \hat{Y}$, then an unbiased estimate of the variance σ^2 can be obtained:

$$\hat{\sigma}^2 = \frac{\hat{\varepsilon}^T \hat{\varepsilon}}{n - p - 1}$$

At the same time, the variance of $\hat{\beta}$ can also be obtained as:

$$var\left(\hat{\beta}\right) = \sigma^2 (X^T X)^{-1}$$

3.2 Full Model Summary Statistics

Build a multiple regression model:

$$Penalties = \beta_0 + \beta_1 Speed + \beta_2 Balance + \beta_3 Strength + \beta_4 Jumping + \beta_5\ Heading + \beta_6 Ball_{Control}$$

The regression results are presented in Table 2. The results reveal that the coefficients of some independent variables pass the significance test, while a few others are not significant with large

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	21730.463	6	3621.744	39.584	.000[b]
	Residual	6221.617	68	91.494		
	Total	27952.080	74			

a. Dependent Variable: Penalties

b. Predictors: (Constant), Ball_Control, Jumping, Strength, Speed, Balance, Heading

P values. The R-square value of the model reached 77.7%, with a high degree of fit.

. Table 2 ANOVA table of the full regression model.

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-3.041	15.925		-.191	.849
	Speed	.085	.126	.056	.670	.505
	Balance	-.262	.130	-.227	-2.008	.049
	Strength	-.004	.169	-.003	-.026	.979
	Jumping	.154	.145	.093	1.057	.294
	Heading	-.238	.111	-.280	-2.150	.035
	Ball_Control	1.087	.133	1.182	8.185	.000

a. Dependent Variable: Penalties

Table 3: Parameter estimates of the Full Regression Model

From the output, we can se that strength is the least significant variables with a p value of .979.

Now we need to remove strength and run the regression model again.

$$Penalties = \beta_0 + \beta_1 Speed + \beta_2 Balance + \beta_3 Jumping + \beta_4\, Heading + \beta_5 Ball_{Control}$$

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	-3.361	10.020		-.335	.738
	Speed	.085	.125	.056	.678	.500
	Balance	-.260	.114	-.225	-2.283	.025
	Jumping	.153	.141	.093	1.081	.283
	Heading	-.239	.097	-.281	-2.453	.017
	Ball_Control	1.088	.132	1.183	8.259	.000

a. Dependent Variable: Penalties

Table 4: Results of the reduced regression model with strength removed

The new model output in Table 4 shows that there are still insignificant variables with Speed being the least significant one having a P value of .50. Next, we are going to remove Speed and run the regression model again:

$$Penalties = \beta_0 + \beta_2 Balance + \beta_3 Jumping + \beta_4\, Heading + \beta_5 Ball_{Control}$$

Table 5: Results of the Regression Model with strength and speed removed

From Table 5 we can see that Jumping is insignificant so we are going to remove it and run the

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	-2.663	9.928		-.268	.789
	Balance	-.242	.110	-.209	-2.194	.032
	Jumping	.189	.130	.115	1.453	.151
	Heading	-.235	.097	-.277	-2.425	.018
	Ball_Control	1.107	.128	1.203	8.640	.000

a. Dependent Variable: Penalties

regression again. The new output of parameter estimates is shown in Table 6.

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	3.064	4.158		.737	.463
	Ball_Control	.798	.053	.868	14.932	.000

a. Dependent Variable: Penalties

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	9.686	5.175		1.872	.065
	Balance	-.195	.106	-.168	-1.833	.071
	Heading	-.138	.071	-.162	-1.949	.055
	Ball_Control	1.002	.107	1.090	9.376	.000

a. Dependent Variable: Penalties

Coefficients[a]

Model		Unstandardized Coefficients B	Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	4.025	4.220		.954	.343
	Heading	-.077	.063	-.090	-1.208	.231
	Ball_Control	.851	.069	.925	12.395	.000

a. Dependent Variable: Penalties

Table 6: Results of the Regression Model with Balance, Heading and Ball-control.

From Table 6 we can see that Balance is now the least significant variable, so we will remove it and run the regression again. The new output is shown in Table 7.

Table 7: Results of the Regression Model with Heading and Ball-control.

From Table 7 we see that Heading is the only insignificant variable now, so we are going to remove it and run the regression with only Ball-Control as independent variable. The new results are shown in Table 8. The results in Table 8 are final. It shows that Ball_Control are the only one variable has real impact penalties. The R-square value is 75.3%.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	21057.684	1	21057.684	222.965	.000[b]
	Residual	6894.396	73	94.444		
	Total	27952.080	74			

a. Dependent Variable: Penalties

b. Predictors: (Constant), Ball_Control

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.868[a]	.753	.750	9.718

a. Predictors: (Constant), Ball_Control

Table 8: Results of the Final Regression Model with only Ball-control.

Thus, our final linear regression model is fitted as:

$$Penalties = 3.064 + 0.798 * Ball_{Control}$$

4.Cross Validation

4.1 Cross Validation

Cross validation randomly selects a certain proportion of data as training samples and the other remaining samples as reserved samples. First, it is necessary to obtain the regression equation on the training samples, and thereafter make predictions on the reserved examples. Since the retained sample does not involve selecting model parameters, the model can obtain a more accurate estimate of its ability of predicting new data. Very often, a 5-fold cross validation is used. That indicates 80% of the data are randomly selected to train a model and predict the reserved 20% of the data.

Here we randomly select 20% of the data, i.e., 15 players from the 75 players as the validation set and used the other 60 players' information to build the regression model. Please see the output in Figure 1. Then we use the model to predict the reserved players' penalties score given their input of Ball-control value. Finally, the predicted value is compared to the true penalty value and a mean squared error is computed.

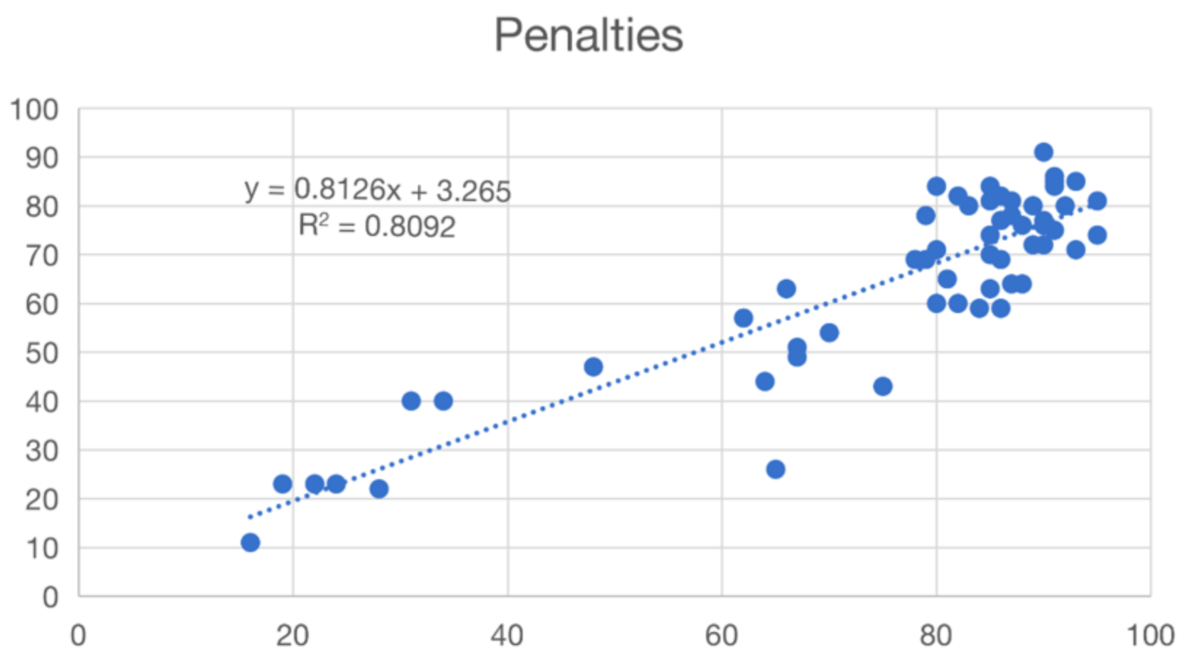


Figure 1 Regression output of the training data.

4.2Prediction

We now use the trained regression model to predict the remaining 20% of the data and show the result in Table 2. The mean squared error is computed as 169.69 and the RMSE is 13.027. Figure 2 plot the predicted values against the true penalties

Name	Ball Control	Penalty	Predicted Value	Squared Error
Arturo Vidal	83	84	70.7108	176.6028366
Antoine Griezmann	86	71	73.1486	4.61648196
Sokratis	58	38	50.3958	153.6558576
Leonardo Bonucci	75	70	64.21	33.5241
Giorgio Chiellini	55	50	47.958	4.169764
Sergio Ramos	83	68	70.7108	7.34843664
Carvajal	83	45	70.7108	661.0452366
Thibaut Courtois	23	27	21.9548	25.45404304
Mesut özil	90	67	76.399	88.341201
Toni Kroos	85	73	72.336	0.440896
Aymeric Laporte	71	35	60.9596	673.9008322
Jérôme Boateng	72	46	61.7722	248.7622928
Diego Godín	76	50	65.0226	225.6785108
Mats Hummels	77	68	65.8352	4.68635904
Lorenzo Insigne	90	61	76.399	237.129201
			Mean Squared Error=	169.690
			RMSE=	13.027

Table 2. Prediction result on the validation set.

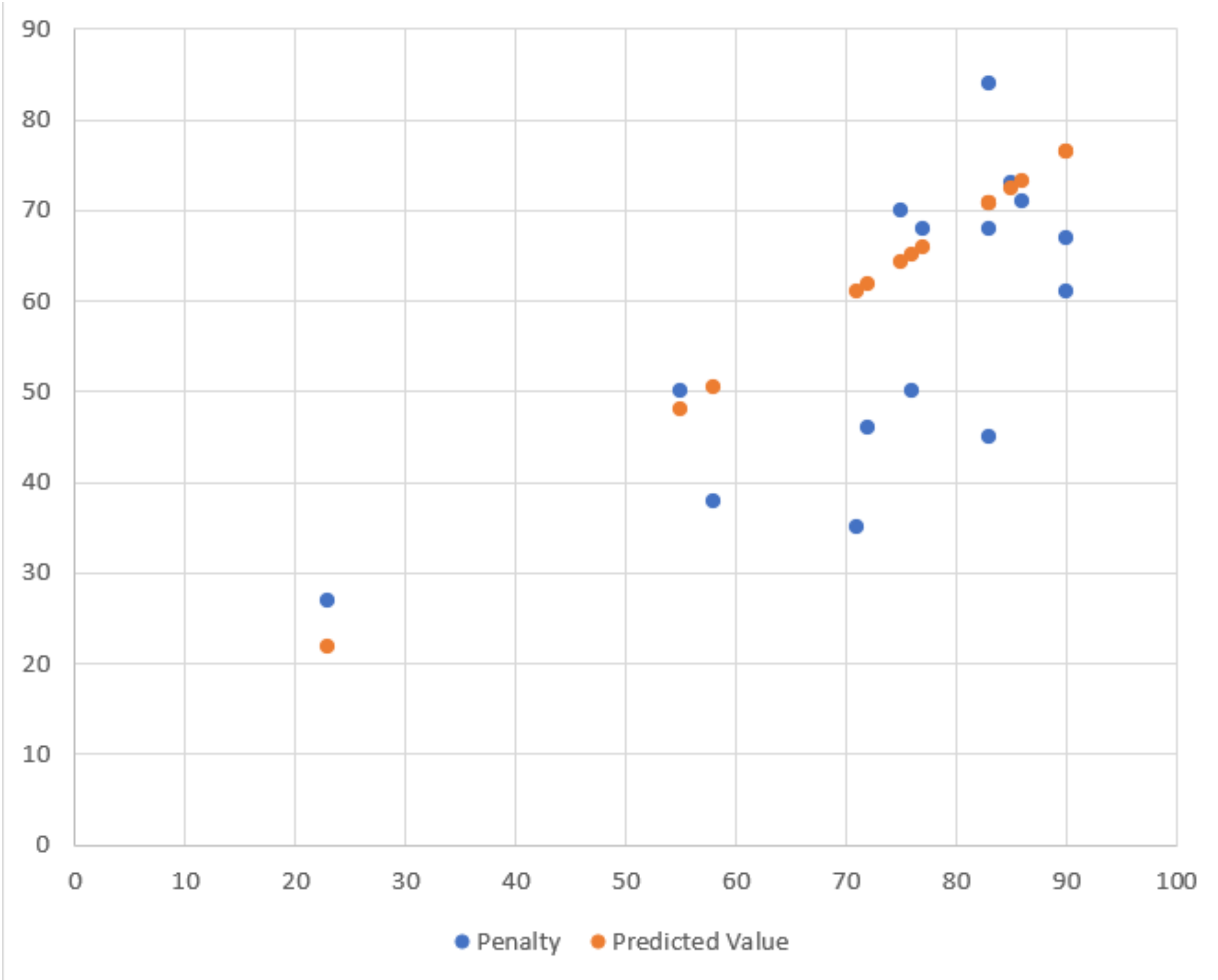


Figure 2. Plot of the predicted values against the true penalties.

5.Model Checking

5.1Model Assumptions

For our regression model, we check the model assumptions:

1. **Normality**: Residuals are normally distributed.
2. **Linearity**: The relationship between X and the mean of Y is linear.
3. **Homoscedasticity**: The variance of residuals is the same for any value of X.
4. **Independence**: residuals are independent of each other.

We plot the distribution of standardized residuals of the dependent variable "Penalties" in Figure 1. It roughly follows a normal distribution with a slight tendency skewing to the left. We also plot the normal P-P plot of residuals in Figure 2 which is basically a straight line.

2. The scatter plot of the dependent variable against the independent variable in Figure 3 shows a linear relationship between Ball_Control and Penalties.

3. The residual plot against fitted values in Figure 4 shows that the error variance is roughly constant. Further we do a non-constant error variance test in R to confirm our conclusion:

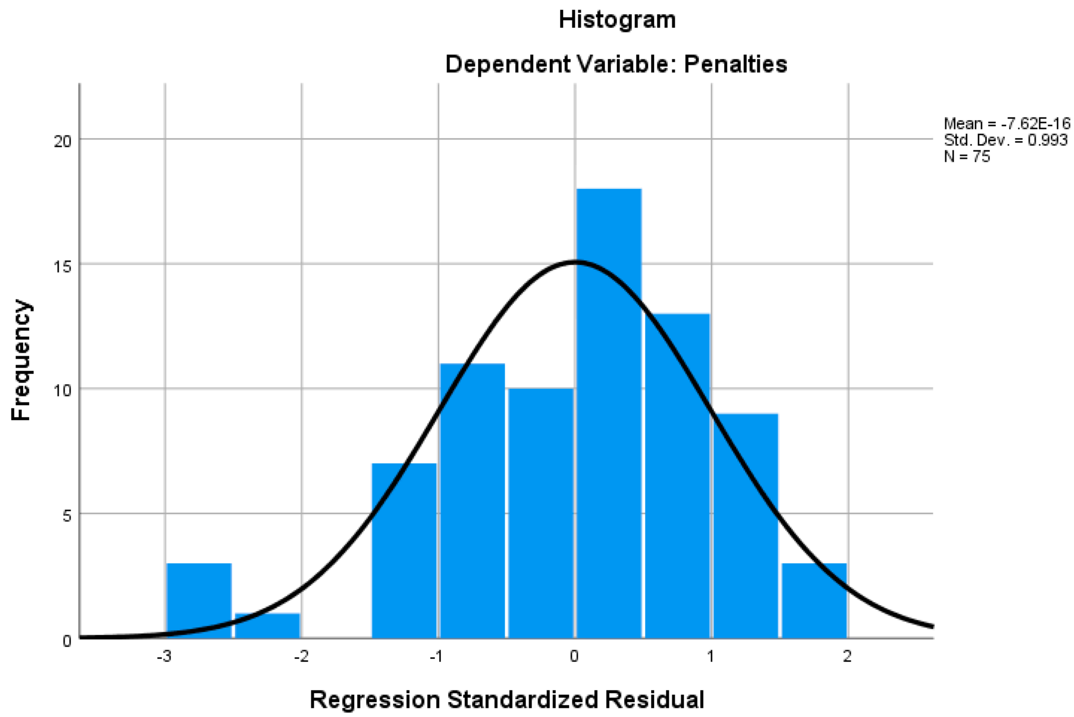
```
> ncvTest(model1)

Non-constant Variance Score Test

Variance formula: ~ fitted.values

Chisquare = 0.08813901, Df = 1, p = 0.76656
```

4. Although there might be a few outliers found in Figure 4, the overall pattern is random and has no serial correlation. So we can conclude that the residuals are independent.


Histogram
Dependent Variable: Penalties
Mean = -7.62E-16
Std. Dev. = 0.993
N = 75
Frequency
20
15
10
5
0
-3
-2
-1
0
1
2
Regression Standardized Residual

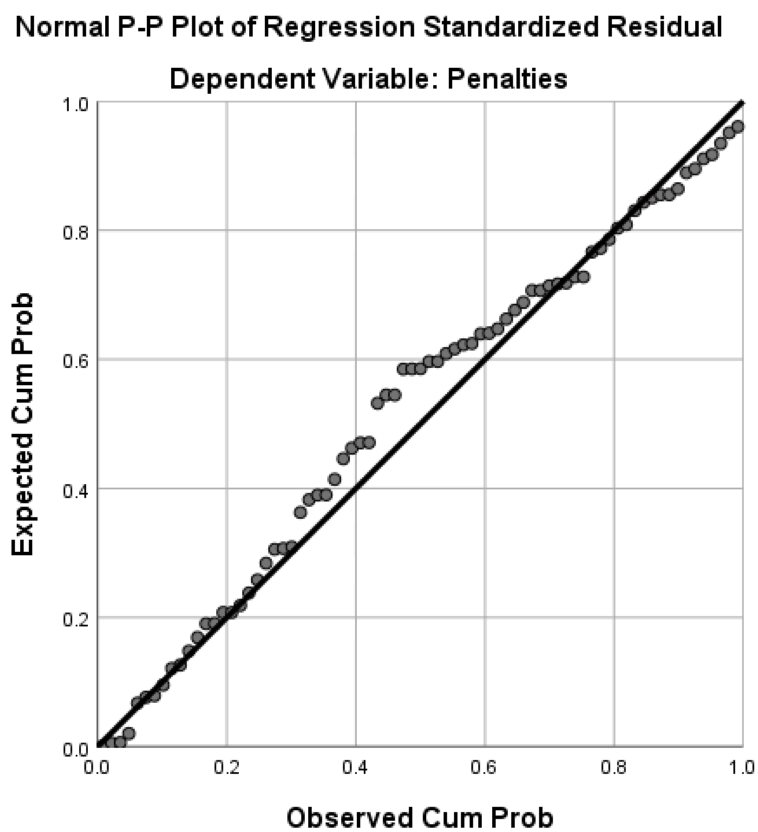
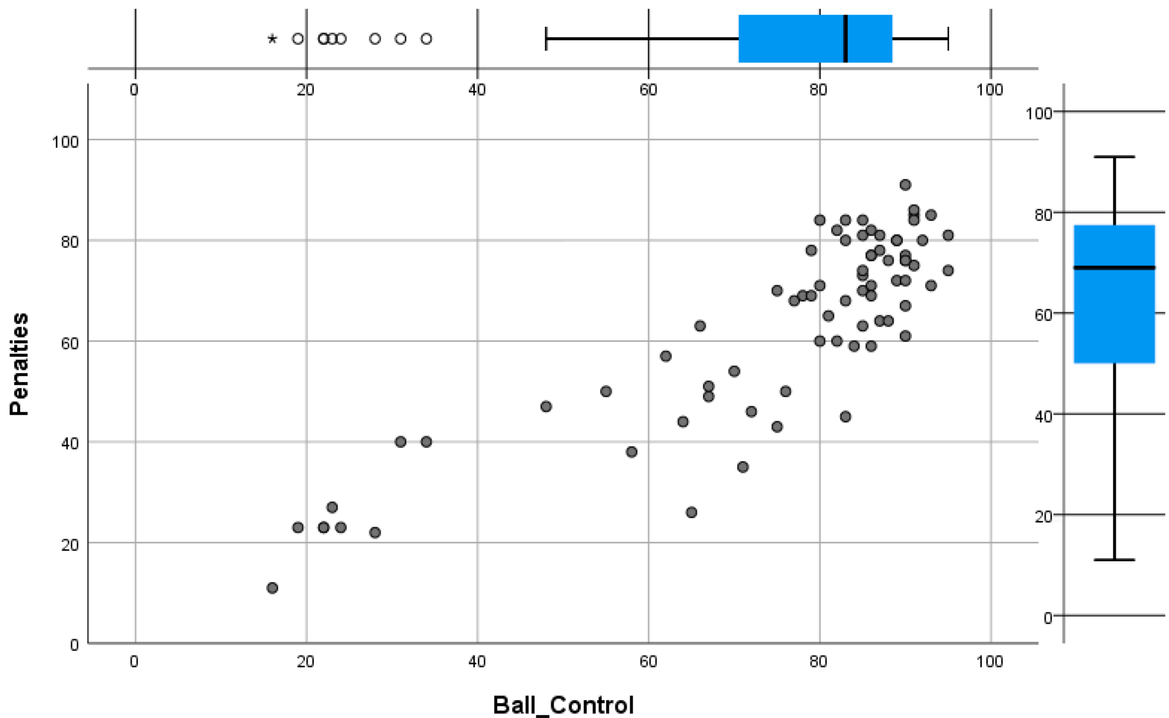

Normal P-P Plot of Regression Standardized Residual
Dependent Variable: Penalties
Expected Cum Prob
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Observed Cum Prob

Figure 1. Histogram of Standardized Residuals

Figure 2 Normal P-P lot of Residuals.

Figure 3 Scatter plot of Penalties against Ball-control



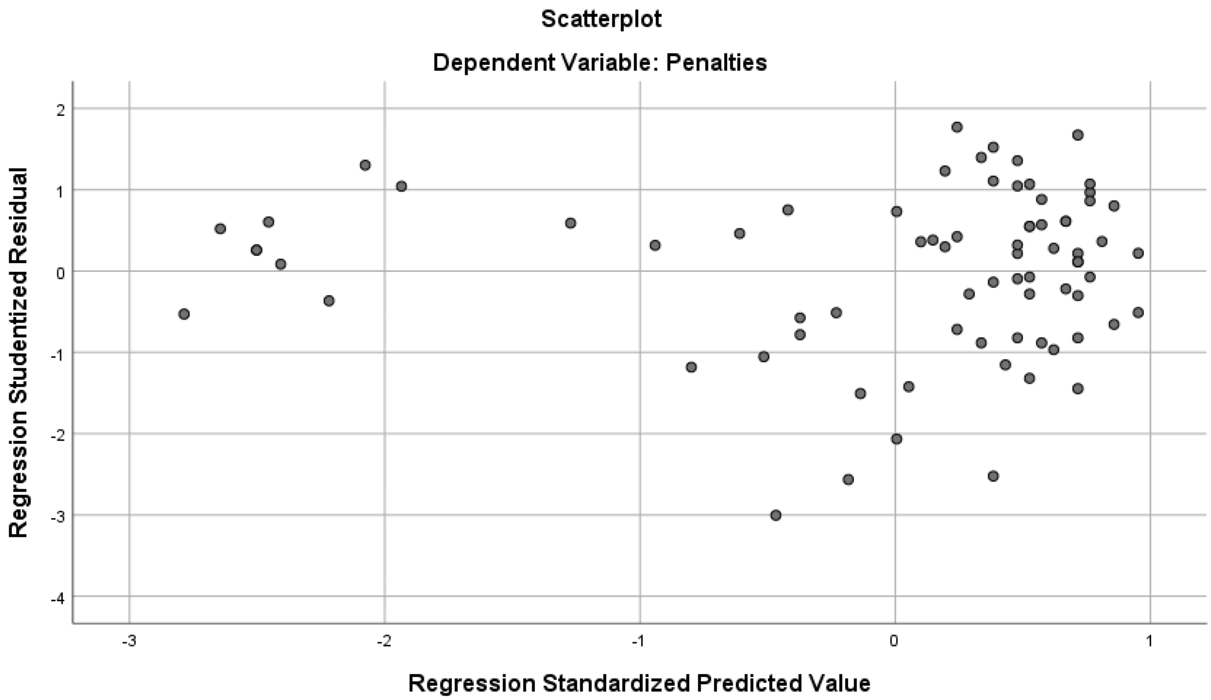


Figure 4 Residual plot against fitted values.

6.Summary and Conclusion

This paper studies the association of soccer player's penalty ability with other abilities such as speed and strength, etc. First, we conducted a descriptive statistical analysis of the independent and dependent variables in the soccer player data set and determined the research objectives. Next, a multiple linear regression model is fitted and a final model is provided after removing insignificant variables. We find that ball-control ability is the most important variable to explain a player's penalty ability. To confirm, we also random select 20% of the data as a validation set to confirm our finding. It demonstrated that the prediction effect is sufficiently accurate. Furthermore, this paper conducted model checking on model assumptions to justify our final result.

Appendix I – Full Data Set

Name	Nationality	National_Position	National_Kit	Club	Club_Position	Club_Kit	Club_Joining	Contract_Expiry	Rating	Height	Weight	Preffered_Foot	Birth_Date	Age	Preffered_Position	Work_Rate
Cristiano Ronaldo	Portugal	LS	7	Real Madrid	LW	7	07/01/2009	2021	94	185 cm	80 kg	Right	02/05/1985	32	LW/ST	High / Low
Lionel Messi	Argentina	RW	10	FC Barcelona	RW	10	07/01/2004	2018	93	170 cm	72 kg	Left	06/24/1987	29	RW	Medium / Medium
Neymar	Brazil	LW	10	FC Barcelona	LW	11	07/01/2013	2021	92	174 cm	68 kg	Right	02/05/1992	25	LW	High / Medium
Luis Suárez	Uruguay	LS	9	FC Barcelona	ST	9	07/11/2014	2021	92	182 cm	85 kg	Right	01/24/1987	30	ST	High / Medium
Manuel Neuer	Germany	GK	1	FC Bayern	GK	1	07/01/2011	2021	92	193 cm	92 kg	Right	03/27/1986	31	GK	Medium / Medium
De Gea	Spain	GK	1	Manchester Utd	GK	1	07/01/2011	2019	90	193 cm	82 kg	Right	11/07/1990	26	GK	Medium / Medium
Robert Lewandowski	Poland	LS	9	FC Bayern	ST	9	07/01/2014	2021	90	185 cm	79 kg	Right	08/21/1988	28	ST	High / Medium
Gareth Bale	Wales	RS	11	Real Madrid	RW	11	09/02/2013	2022	90	183 cm	74 kg	Left	07/16/1989	27	RW	High / Medium
Zlatan Ibrahimovi膰	Sweden			Manchester Utd	ST	9	07/01/2016	2017	90	195 cm	95 kg	Right	10/03/1981	35	ST	Medium / Low
Thibaut Courtois	Belgium	GK	1	Chelsea	GK	13	07/26/2011	2019	89	199 cm	91 kg	Left	05/11/1992	24	GK	Medium / Medium
Jérôme Boateng	Germany	RCB	17	FC Bayern	Sub	17	07/14/2011	2021	89	192 cm	90 kg	Right	09/03/1988	28	CB	Medium / Medium
Eden Hazard	Belgium	LF	10	Chelsea	LW	10	07/01/2012	2020	89	173 cm	74 kg	Right	01/07/1991	26	LW/LM	High / Medium
Luka Modrić	Croatia			Real Madrid	RCM	19	08/01/2012	2020	89	174 cm	65 kg	Right	09/09/1985	31	CM/CDM	High / Medium
Mesut özil	Germany	CAM	10	Arsenal	CAM	11	09/02/2013	2018	89	180 cm	76 kg	Left	10/15/1988	28	CAM/LW	Medium / Low
Gonzalo Higuaín	Argentina	Sub	9	Juventus	ST	9	07/26/2016	2021	89	184 cm	92 kg	Right	12/10/1987	29	ST	High / Medium
Thiago Silva	Brazil	Sub	14	PSG	LCB	2	07/01/2012	2020	89	183 cm	79 kg	Right	09/22/1984	32	CB	High / High
Sergio Ramos	Spain	LCB	15	Real Madrid	LCB	4	08/01/2005	2020	89	183 cm	75 kg	Right	03/30/1986	31	CB	High / Medium
Sergio Agüero	Argentina	Sub	7	Manchester City	ST	10	07/28/2011	2020	89	173 cm	70 kg	Right	06/02/1988	28	ST	High / Medium
Paul Pogba	France			Manchester Utd	LCM	6	08/09/2016	2021	88	191 cm	84 kg	Right	03/15/1993	24	CM/CAM	High / Medium
Antoine Griezmann	France	CAM	7	Atl茅tico Madrid	RS	7	07/29/2014	2021	88	176 cm	67 kg	Left	03/21/1991	26	ST/LW	High / Medium
Kevin De Bruyne	Belgium	RCM	7	Manchester City	RCM	17	08/30/2015	2021	88	181 cm	68 kg	Right	06/28/1991	25	CAM/RM/LM	High / High
Marco Reus	Germany			Bor. Dortmund	LW	11	07/01/2012	2019	88	182 cm	76 kg	Right	05/31/1989	27	LM/CAM	Medium / Medium
Alexis S谩nchez	Chile	LW	7	Arsenal	ST	7	07/10/2014	2018	88	169 cm	62 kg	Right	12/19/1988	28	ST/RM	High / High
Toni Kroos	Germany	LDM	8	Real Madrid	LCM	8	07/17/2014	2022	88	182 cm	78 kg	Right	01/04/1990	27	CM/CDM	Medium / Medium
Diego Godín	Uruguay	LCB	3	Atl茅tico Madrid	LCB	2	08/01/2010	2019	88	185 cm	73 kg	Right	02/16/1986	31	CB	Medium / High
Mats Hummels	Germany	LCB	5	FC Bayern	LCB	5	07/01/2016	2021	88	191 cm	92 kg	Right	12/16/1988	28	CB	High / Medium
Hugo Lloris	France	GK	1	Spurs	GK	1	08/01/2012	2022	88	188 cm	82 kg	Left	12/26/1986	30	GK	Medium / Medium
Giorgio Chiellini	Italy	LCB	3	Juventus	LCB	3	07/01/2005	2018	88	187 cm	84 kg	Left	08/14/1984	32	CB	Low / High
Philipp Lahm	Germany			FC Bayern	RB	21	11/01/2002	2018	88	170 cm	66 kg	Right	11/11/1983	33	RB/CM	Medium / High
Pepe	Portugal	RCB	3	Real Madrid	RCB	3	07/01/2007	2017	88	188 cm	81 kg	Right	02/26/1983	34	CB	Medium / High
Petr čech	Czech Republic			Arsenal	GK	33	06/29/2015	2019	88	196 cm	90 kg	Left	05/20/1982	34	GK	Medium / Medium
Gianluigi Buffon	Italy	GK	1	Juventus	GK	1	07/01/2001	2018	88	192 cm	91 kg	Right	01/28/1978	39	GK	Medium / Medium
Iniesta	Spain	LCM	6	FC Barcelona	LM	8	07/01/2002	2018	88	171 cm	68 kg	Right	05/11/1984	32	CM	High / Medium
Jan Oblak	Slovenia	GK	1	Atl茅tico Madrid	GK	13	07/17/2014	2021	87	189 cm	87 kg	Right	01/07/1993	24	GK	Medium / Medium
James Rodríguez	Colombia	LAM	10	Real Madrid	CAM	10	07/22/2014	2020	87	180 cm	75 kg	Left	07/12/1991	25	CAM/RM	Medium / Low
Pierre-Emerick Aubameyang	Gabon			Bor. Dortmund	ST	17	07/04/2013	2020	87	187 cm	80 kg	Right	06/18/1989	27	ST/RM	Medium / Low
Leonardo Bonucci	Italy	CB	19	Juventus	RCB	19	07/01/2010	2021	87	190 cm	85 kg	Right	05/01/1987	29	CB	Low / High
Arturo Vidal	Chile	CDM	8	FC Bayern	LDM	23	07/28/2015	2019	87	180 cm	75 kg	Right	05/22/1987	29	CM/CDM	High / High
Ivan Rakitić	Croatia			FC Barcelona	CAM	4	06/24/2014	2021	87	184 cm	78 kg	Right	03/10/1988	29	CM	Medium / Medium
David Silva	Spain	LM	21	Manchester City	LCM	21	07/14/2010	2019	87	173 cm	67 kg	Left	01/08/1986	31	CAM/LM	High / Low
Samir Handanovi膰	Slovenia			Inter	GK	1	07/01/2012	2019	87	193 cm	89 kg	Right	07/14/1984	32	GK	Medium / Medium
Piqué	Spain	RCB	3	FC Barcelona	RCB	3	07/01/2008	2019	87	193 cm	85 kg	Right	02/02/1987	30	CB	High / Medium
Arjen Robben	Netherlands	RW	11	FC Bayern	RM	10	08/28/2009	2018	87	180 cm	80 kg	Left	01/23/1984	33	RM/RW	High / Low
Paulo Dybala	Argentina	Sub	21	Juventus	CAM	21	07/01/2015	2020	86	177 cm	74 kg	Left	11/15/1993	23	ST/CAM	Medium / Medium
Marco Verratti	Italy	LCM	10	PSG	RCM	6	07/01/2012	2021	86	165 cm	60 kg	Right	11/05/1992	24	CM/CDM	High / High
David Alaba	Austria	LM	8	FC Bayern	LB	27	02/10/2010	2021	86	180 cm	76 kg	Left	06/24/1992	24	LB/CM	High / Medium
Henrikh Mkhitaryan	Armenia			Manchester Utd	RW	22	07/06/2016	2020	86	177 cm	75 kg	Right	01/21/1989	28	RM/CAM	High / High
Bernd Leno	Germany	Sub	12	Bayer 04	GK	1	01/01/2012	2018	86	190 cm	83 kg	Right	03/04/1992	25	GK	Medium / Medium
Thomas Müller	Germany	RM	13	FC Bayern	CAM	25	08/10/2008	2021	86	186 cm	75 kg	Right	09/13/1989	27	CF/CAM/ST	High / High
Sergio Busquets	Spain	CDM	5	FC Barcelona	CDM	5	09/01/2008	2021	86	189 cm	76 kg	Right	07/16/1988	28	CDM/CM	Medium / Medium
Thiago	Spain	Sub	10	FC Bayern	RDM	6	07/14/2013	2019	86	174 cm	70 kg	Right	04/11/1991	25	CM/CDM/LM	Medium / Medium
Jordi Alba	Spain	LB	18	FC Barcelona	LCB	18	07/01/2012	2020	86	170 cm	68 kg	Left	03/21/1989	28	LB	High / Medium
Coutinho	Brazil	RW	11	Liverpool	LW	10	01/30/2013	2022	86	171 cm	68 kg	Right	06/12/1992	24	LW/CAM	High / High
Edinson Cavani	Uruguay	RS	21	PSG	ST	9	07/16/2013	2018	86	184 cm	71 kg	Right	02/14/1987	30	ST	High / High
Radja Nainggolan	Belgium	Sub	16	Roma	LF	4	01/07/2014	2020	86	176 cm	65 kg	Right	05/04/1988	28	CM/CAM	High / High
Miranda	Brazil	RCB	3	Inter	RCB	25	07/03/2015	2018	86	186 cm	78 kg	Right	09/07/1984	32	CB	Medium / Medium
Karim Benzema	France			Real Madrid	ST	9	07/01/2009	2019	86	187 cm	79 kg	Right	12/19/1987	29	ST	Medium / Low
Sokratis	Greece	RCB	19	Bor. Dortmund	CB	25	07/01/2013	2019	85	186 cm	85 kg	Right	06/09/1988	28	CB	Medium / Medium
Laurent Koscielny	France	RCB	21	Arsenal	LCB	6	07/07/2010	2020	85	186 cm	75 kg	Right	09/10/1985	31	CB	Medium / High
Filipe Lu铆s	Brazil	Sub	6	Atl茅tico Madrid	LB	3	07/29/2015	2019	85	182 cm	77 kg	Left	08/09/1985	31	LB	High / Medium
Vincent Kompany	Belgium	CB	3	Manchester City	Sub	4	08/22/2008	2019	85	193 cm	85 kg	Right	04/10/1986	30	CB	Medium / Medium
Aymeric Laporte	France			Athletic Bilbao	LCB	4	10/01/2012	2020	84	189 cm	85 kg	Left	05/27/1994	22	CB	Medium / Medium
Yannick Carrasco	Belgium	LM	18	Atl茅tico Madrid	LM	10	07/10/2015	2022	84	180 cm	66 kg	Right	09/04/1993	23	LM/LW	High / Medium
Carvajal	Spain	RB	12	Real Madrid	RB	2	07/05/2013	2020	84	173 cm	73 kg	Right	01/11/1992	25	RB	High / Medium
Riyad Mahrez	Algeria			Leicester City	RM	26	01/11/2014	2020	84	179 cm	62 kg	Left	02/21/1991	26	RM/CAM	Medium / Medium
Rapha毛l Varane	France			Real Madrid	Sub	5	07/01/2011	2020	84	191 cm	78 kg	Right	04/25/1993	23	CB	Medium / High
Mauro Icardi	Argentina			Inter	ST	9	07/09/2013	2021	84	181 cm	75 kg	Right	02/19/1993	24	ST	Medium / Low
Lorenzo Insigne	Italy	Sub	20	Napoli	LW	24	07/01/2010	2019	84	163 cm	59 kg	Right	06/04/1991	25	LW	High / Medium
Isco	Spain	Sub	22	Real Madrid	Sub	22	07/03/2013	2018	84	176 cm	74 kg	Right	04/21/1992	24	CM/RM	High / Medium
Koke	Spain	RCM	8	Atl茅tico Madrid	RM	6	01/01/2011	2019	84	178 cm	74 kg	Right	01/08/1992	25	CM/RM	High / High
Kostas Manolas	Greece	LCB	4	Roma	RCB	44	08/26/2014	2019	84	189 cm	83 kg	Right	06/14/1991	25	CB	Medium / High
Danijel Suba拧i膰	Croatia			AS Monaco	GK	1	01/24/2012	2019	84	191 cm	84 kg	Right	10/27/1984	32	GK	Medium / Medium
Nicol谩s Otamendi	Argentina	RCB	17	Manchester City	LCB	30	08/20/2015	2020	84	183 cm	81 kg	Right	02/12/1988	29	CB	High / High
Shkodran Mustafi	Germany	Sub	2	Arsenal	RCB	20	08/30/2016	2021	84	184 cm	82 kg	Right	04/17/1992	24	CB	Medium / High
Nemanja Mati膰	Serbia			Chelsea	LCM	21	01/15/2014	2019	84	194 cm	84 kg	Left	08/01/1988	28	CDM/CM	Medium / High

4	5	93	92	22	23	31	63	96	94	29	85	86	84	83	77	91	92	92	80	63	90	95	85	92	93	90	81	76	85	88
4	4	95	97	13	26	28	48	95	93	22	90	94	77	88	87	92	87	74	59	95	90	68	71	85	95	88	89	90	74	85
5	5	95	96	21	33	24	56	88	90	36	80	80	75	81	75	93	90	79	49	82	96	61	62	78	89	77	79	84	81	83
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4	1	48	30	10	11	10	29	85	12	30	70	70	15	55	59	58	61	44	83	35	52	78	25	25	13	16	14	11	47	11
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3	1	23	13	11	16	18	23	81	13	15	44	52	14	32	31	46	52	38	70	45	61	68	13	36	14	17	19	11	27	12
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4	4	91	93	25	22	27	54	85	85	41	86	86	80	84	80	93	87	77	65	90	92	59	57	79	81	82	82	79	86	79
4	4	92	86	66	73	80	62	88	79	76	90	76	78	92	83	77	71	83	58	94	93	67	55	73	71	82	79	77	80	74
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4	3	85	84	12	18	22	50	86	92	20	70	84	68	73	59	79	81	73	85	69	75	77	80	86	92	80	74	62	70	88
3	3	80	68	90	89	91	77	84	59	91	74	80	60	79	81	72	76	74	81	68	75	93	82	78	38	71	61	73	71	63
3	3	83	61	85	90	89	84	82	52	88	63	76	66	76	70	77	78	84	81	60	80	92	90	79	60	55	73	67	68	66
4	4	89	89	13	12	20	57	88	91	24	83	90	70	79	63	92	86	74	73	90	86	80	68	87	90	84	82	72	80	85
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3	4	86	87	23	11	22	69	90	91	35	76	81	82	79	75	87	86	75	61	80	90	87	80	81	88	82	84	75	71	87
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3	4	86	89	30	35	39	80	87	86	42	82	82	80	80	73	88	84	85	72	87	90	85	70	84	85	81	78	78	77	83
5	3	85	78	63	65	75	60	85	76	77	88	85	85	89	93	55	37	78	74	62	70	34	54	87	75	88	85	84	73	82
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3	1	16	12	14	18	12	34	84	11	19	55	55	13	29	26	43	60	41	78	49	67	76	15	22	11	12	13	14	11	13
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91	89	95	90	89
86	88	87	85	90
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85	85	76	86	83
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12	6	5	8	5
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10	9	9	11	10
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7	13	5	9	9
9	8	14	4	10
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10	14	5	12	13
15	8	8	6	12
83	82	79	79	84
11	12	8	5	12
10	11	15	9	6

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